Higher National Diploma in Engineering (Electrical)
Fourth Year, First Semester Examination – 2019
EE 3206 – Robotics and Automation

Instructions for Candidates:
answer **any Five (05)** questions.
All question carry equal marks

No. of questions: 06
No. of pages : 03
Time : **03 Hours**

Q1.

i) Briefly explain what Surface Mounting Technology is? **(4 marks)**

ii) Explain the Printed Circuit Board (PCB)'s manufacturing Process flow chart by including the following key points. **(8 marks)**

- Print & Etch
- Drilling
- Routing
- Deburring
- Wave & flow soldering

iii) A pump that fails three times over the span of a workday. The time spent repairing each of those breakdowns took to totally one hour.
Calculate MTTR (Mean Time To Repair)? **(3 marks)**

iv) Briefly explain the Solid State Devices and state any two advantages of Solid state devices? **(5 marks)**

Q2.

i) What is Forward Kinematics and where can it be applied? **(6 marks)**

ii) Briefly describe meaning of DOF? **(4 marks)**

iii) Calculate the number of DOF for kinematic system shown in **Figure no Q2.1**, (Page no-03) by using Grubler's equation. **(4 marks)**

iv) Consider the three degree-of-freedom planar robot arm shown in **Figure no Q2.2**, (Page no-03) which consists of one fixed link (link 0) and three movable links that move on the plane. All the links are connected by revolute joints whose joint axes are all perpendicular to the plane of the links. Derive the equations for end-effector position (x_e,y_e) and orientation (angle,ϕ_e) in terms of joint displacements and link lengths. **(6 marks)**

Q3.

i) Rotary encoders are manufactured in two basic forms. briefly describe rotary encoder forms? **(4 marks)**

ii) Briefly describe the following terms by using suitable examples.
- Sensor
- Actuator
- Transducer
- Sensitivity
- Resolution **(8 marks)**

iii) What are the classifications of Sensor? State **four** Suitable sensors for each one. **(8 marks)**

Q4.

i) What do you mean by Automation and Write the benefits of industrial Automation? **(4 marks)**

ii) What are the types of Robot Programming Methods? **(4 marks)**

iii) Briefly explain online Programming Methods and its applications? **(8 marks)**

iv) The gearing of a machine tool is shown in **Figure no Q4** (Page no-03). The motor shaft is connected to gear A and rotates at 1950 R.P.M. The gear wheels B, C, D and E are fixed to parallel shafts rotating together. The final gear F is fixed on the output shaft. What is the speed of gear F? the number of teeth on each gear are as given below: no. of teeth on each gear,
$T_A = 20$, $T_B = 50$, $T_C = 25$, $T_D = 75$, $T_E = 26$, $T_F = 65$ **(4 marks)**

Q5.

i) What are the main types of Robot Configurations? **(4marks)**

ii) Briefly describe about SCARA Robot Configuration and draw the Workspace of the SCARA robot? **(4 marks)**

iii) Briefly describe the meaning of Repeatability and Accuracy of an Industrial Robot using Suitable Sketch.? **(6 marks)**

iv) Clearly state the basic Pneumatic System Components by using suitable sketch and List out the Advantages of Pneumatic System? **(6 marks)**

Q6.

i) What are the types of Grippers? **(4 marks)**

ii) Clearly describe four ways of grippers mechanism based on kinematic devices by using suitable sketches? **(8 marks)**

iii) Clearly describe the different types of robot work cell layouts used in industry? **(8 marks)**

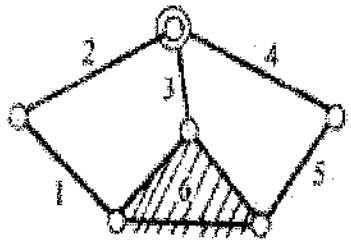

Figure No Q2.1

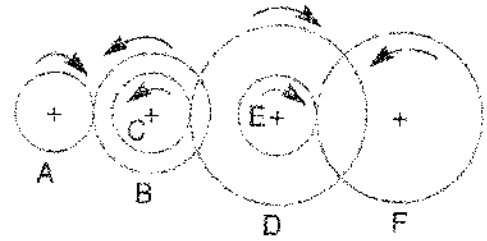

Figure No Q4

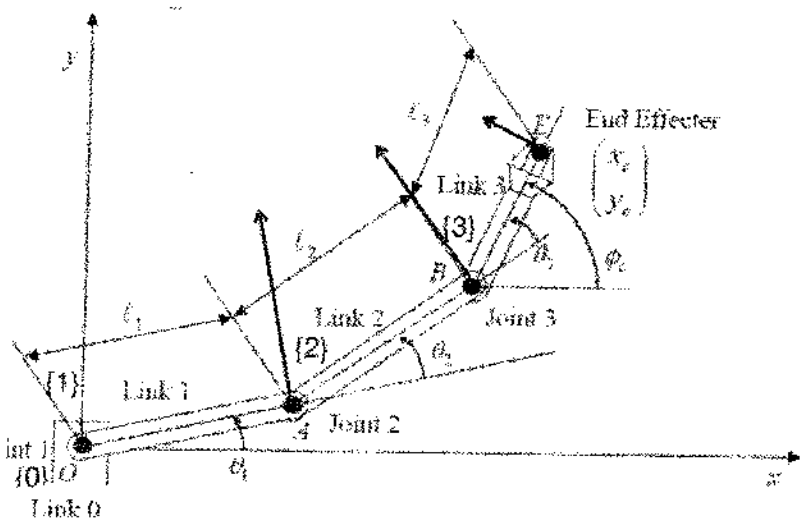

Figure No Q2.2